

Fangyuan Shi

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EDUCATION

M.S. in Operations Research

Aug 2025 – Present

Georgia Institute of Technology, Atlanta, Georgia, USA

GPA: 3.87/4.0

Relevant Coursework: Computational Data Analysis, Deterministic Optimization, Probabilistic Models, Simulation, Nonparametric Data Analysis, Regression Analysis, Time Series Analysis, Economic Decision Analysis

B.Eng. in Industrial Engineering

Sep 2021 – Jun 2025

Sichuan University, Chengdu, China

GPA: 3.89/4.0

Relevant Coursework: Probabilistic Methods in Operations Research, Statistical Testing and Regression, Design of Experiments and Quality Assurance

RESEARCH INTEREST

- Data-driven decision-making in operational systems

ACADEMIC EXPERIENCE

Wildfire Spread Modeling from Satellite Imagery

Apr 2026 – Present

- Developed a pixel-to-temperature conversion pipeline for GOES-17 infrared video (Channel 07), replacing a flawed greyscale method with a colormap-inversion LUT pipeline
- Implemented and debugged a PDE-based numerical simulation of fire spread, diagnosing and correcting a time-scale mismatch that had caused the physics-based model to underperform a naïve baseline

Modeling and Forecasting of Henry Hub Natural Gas Spot Prices

Apr 2026 – Jun 2026

- Developed and compared time-series and machine-learning models (ARIMA/SARIMA, ARIMAX/SARIMAX, ARMA-GARCH, XGBoost, LSTM, Silverkite) for short-term forecasting of Henry Hub natural gas spot prices, incorporating U.S. working-gas storage and heating degree days (HDD) as exogenous variables
- Analyzed dynamic price behaviors driven by supply-demand fluctuations, weather shocks, and market volatility, and evaluated model performance under fixed multi-step and rolling one-step-ahead forecasting frameworks

Expectation–Experience Gap Modeling for Hotel Reviews

Jan 2026 – Jun 2026

Advised by Prof. Xinxin Wang

- Formulated an expectation–experience gap modeling framework for hotel reviews, decomposing user feedback into aspect coverage, sentiment polarity, and helpfulness weights to prioritize service dimensions
- Designed a multimodal modeling pipeline to extract structured representations from platform promotional content (images/text) and customer reviews, using Transformer-based encoders to project content into a unified K-aspect space (e.g., room, service, breakfast, location)
- Completed system design and feature specification; currently implementing aspect-level modeling with planned optimization for gap estimation and ranking

ADDITIONAL INFORMATION

Language Skills: Chinese (Native); English (TOEFL: 101; GRE: 328)

Computer Skills: Python, R, MATLAB